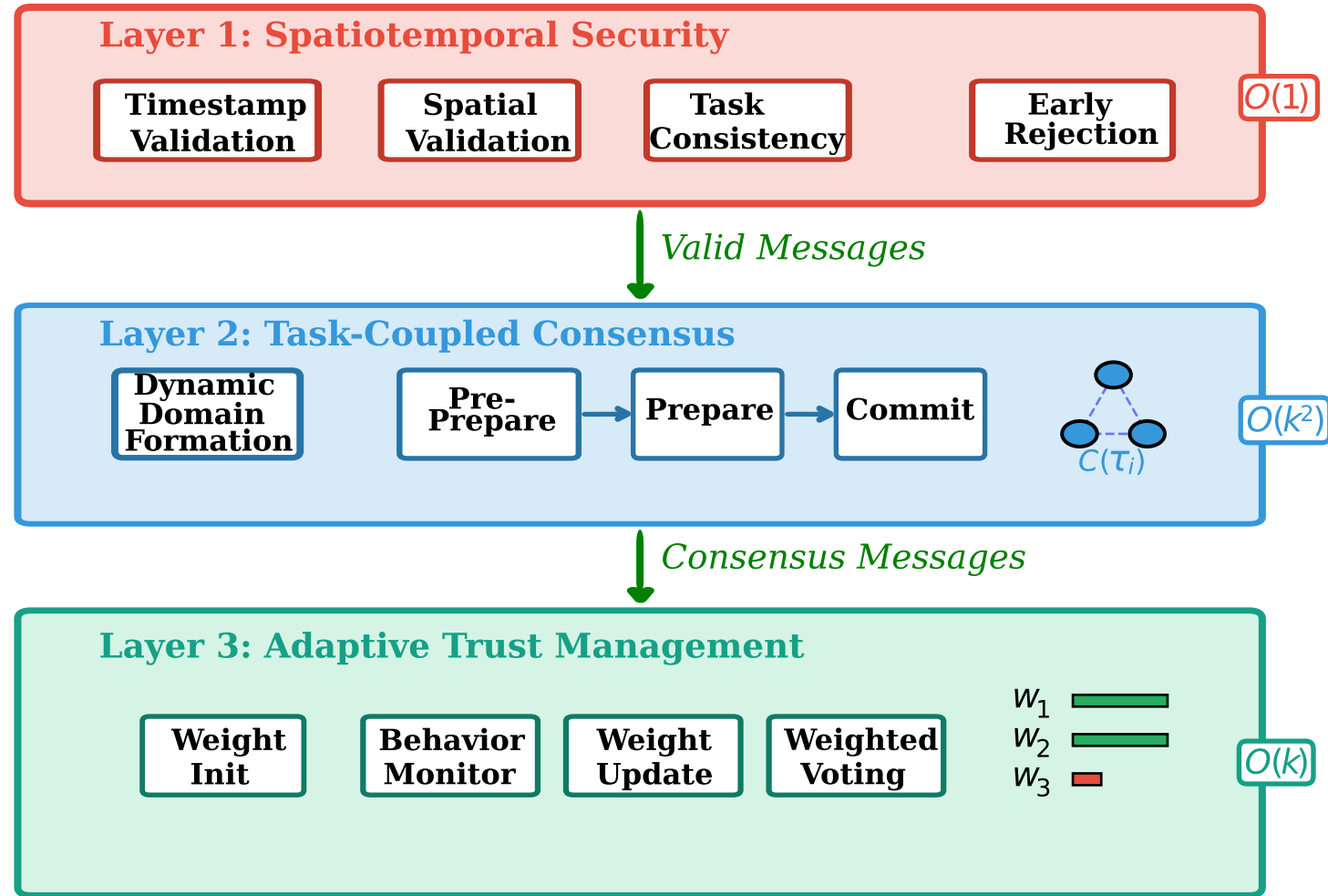
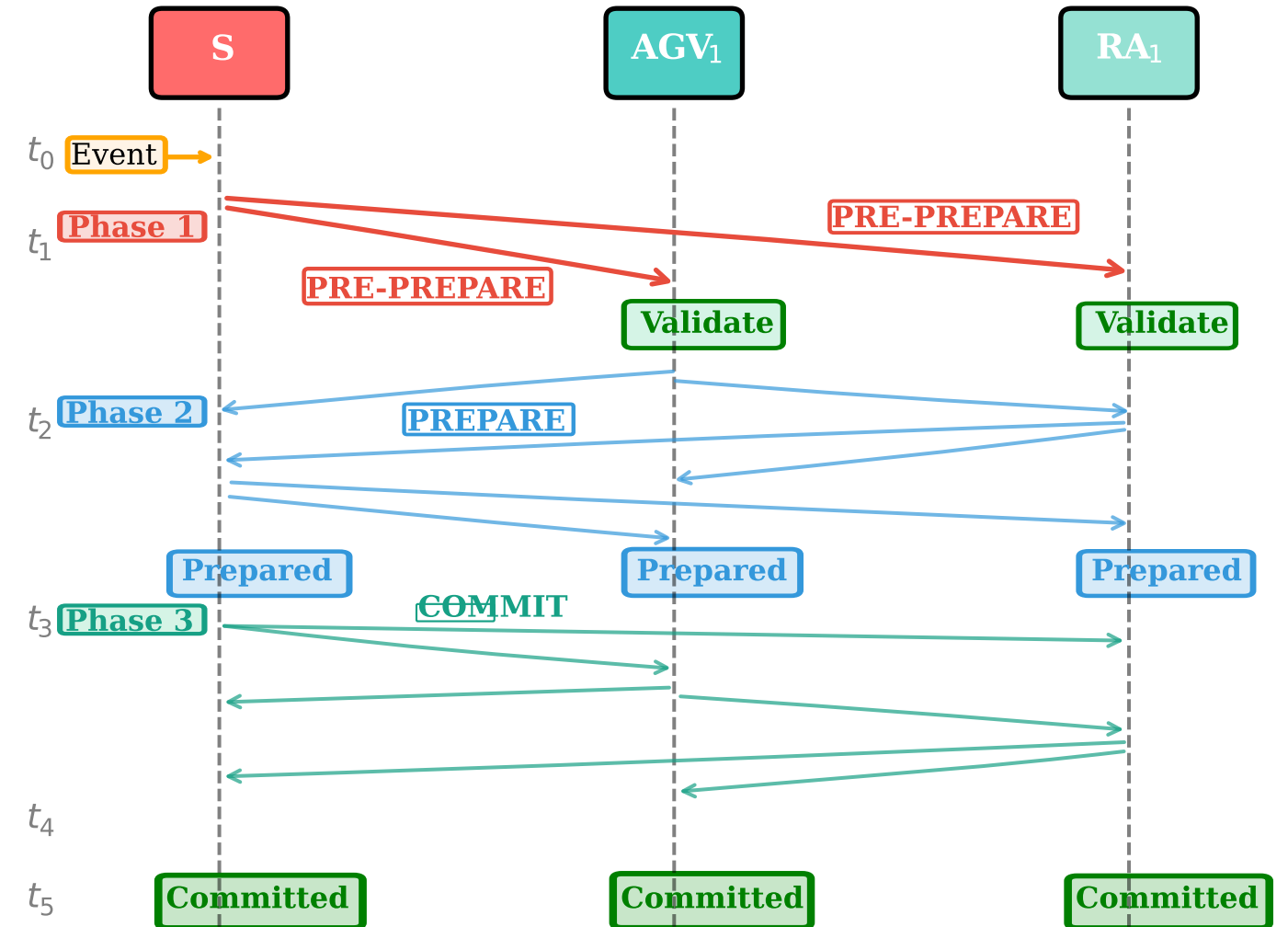


(a) Three-Layer Architecture



$k=3$ task-coupled nodes, $m=3$ scheduler replicas
 Total: $O(k^2+m^2+mk)=9+9+9=27$ messages

(b) Consensus Protocol Message Flow



Agent Consensus: $O(k) = 9$ messages (3 phases)
 Total (with scheduler): $k^2+m^2+mk = 27$ messages
 vs PBFT: $2n^2-n=1,770$ messages (98.5% reduction)